📝 Typed Questions (As it is)

1. What are the different factors to be considered while selecting the site for hydroelectric power plant?

2. Explain the various environmental and non-conventional sources of energy and their role in meeting the power demand in India.

3. What factors must be considered when selecting a nuclear power plant? Discuss each factor in detail.

4. Explain primary and secondary transmission lines.

5. Explain hydroelectric power plant with its advantages and disadvantages.

6. Write the general layout of hydro power plant, explain its components and mention the working of it.

7. Give the layout of diesel power plant.

8. Draw a schematic diagram of a nuclear power plant and explain the function of its major components such as the reactor, turbine and cooling system.

9. What are the advantages and disadvantages of gas turbine power plant? Also discuss the environmental impact of thermal power plants in terms of efficiency, fuel consumption and impact.

10. What are the different factors to be considered while selecting the site for nuclear power plant?

11. Explain nuclear power plant and write its construction and working of nuclear power plant.

12. Draw a schematic diagram of main diesel generating unit and mention its advantages or disadvantages.

13. Explain AC transmission and distribution system.
14. What are the different factors to be considered while selecting the site for thermal power plant? Discuss each factor in detail.
15. Write the environmental impact of thermal power plant.
16. Write the environmental impact of hydro power plants.
17. Explain in detail power plant working. Advantages and disadvantages of thermal power plant.

📝 Typed Questions (Unit – 2: Nuclear, Diesel & Gas Turbine Power Plants)

1. Give the layout of diesel power plant.
2. Draw a schematic diagram of a nuclear power plant and explain the function of its major components such as the reactor system, generator, turbine and cooling system.
3. What are the advantages and disadvantages of gas turbine power plants? Compare them with other conventional power plants in terms of efficiency, fuel consumption and environmental impact.
4. What are the different factors to be considered while selecting the site for nuclear power plant?
5. Explain nuclear power plant. Write its construction and working of nuclear power plant.
6. Draw a schematic diagram of diesel generator unit and main components, advantages or disadvantages.

- **Q1.** Explain the **Photovoltaic Effect** and how it is used to generate electricity in a Solar PV system.

- **Q2.** Describe the **working principle of the Photovoltaic Effect** with suitable examples and a neat block diagram.

- **Q3.** Explain the **construction of a Solar PV panel and PV array** with a proper block diagram.

- **Q4.** Differentiate between **Stand-alone and Grid-connected Solar PV systems** with diagrams.

- **Q5.** Explain **Solar PV systems** in detail. Also write their **advantages and disadvantages** with a block diagram.

- **Q6.** Discuss the **environmental impact of Solar PV systems.**

- **Q7.** Explain the concept of **Global Wind and Local Wind** with examples.

✨ Important Questions – Biomass, Biogas & Fuel Cells 🌱⚡

🔹 **Q1.** Describe the **different types of biogas plants**. Explain their **construction, working principle, and advantages.**

🔹 **Q2.** Explain the **block diagram of Municipal Solid Waste (MSW) to Energy (Incineration Plant).**

🔹 **Q3.** Describe the **Biomass Power Plant.** Also explain the **factors to be considered for site selection.**

🔹 **Q4.** Write the **importance and scope of Biomass Energy.**

🔹 **Q5.** Define **Fuel Cells.** Explain their **classification, working principle, and applications.**

✨ Important Questions – Biomass, Wind & Fuel Cells 🌱⚡

🔹 **Q1.** Write the **factors affecting the distribution of wind energy on the earth's surface.** Also explain the **importance of wind energy.**

🔹 **Q2.** Describe the **different types of biogas plants.** Explain their **construction, working principle, and advantages.**

🔹 **Q3.** Explain the **block diagram of Municipal Solid Waste (MSW) to Energy (Incineration Plant).**

🔹 **Q4.** Describe the **Biomass Power Plant.** Also explain the **factors to be considered for site selection.**

🔹 **Q5.** Write the **importance and scope of Biomass Energy.**

🔹 **Q6.** Define **Fuel Cells.** Explain their **classification, working principle, and applications.**

✨ Important Questions – Biomass, Hybrid Systems & Power Economics ⚡🌱

🔹 **Q1.** Define **Biogas Plants** and explain the **types of biogas plants.**

🔹 **Q2.** Explain the **process of electricity generation using biomass.**

🔹 **Q3.** Explain the **PV – Wind Hybrid System** and its **types.**

🔹 **Q4.** Differentiate between **PV – Wind Hybrid System** and **PV – Fuel Cell Hybrid System** with a suitable **block diagram.**

📘 UNIT – 5: Economics of Power Generation

🔹 **Q5.** Explain the terms:

- Connected Load
- Firm Power
- Cold Reserve
- Hot Reserve
- Spinning Reserve

🔹 **Q4.** Differentiate between **PV – Wind Hybrid System** and **PV – Fuel Cell Hybrid System** with a suitable **block diagram.**

📘 UNIT – 5: Economics of Power Generation

🔹 **Q5.** Explain the terms:

- Connected Load
- Firm Power
- Cold Reserve
- Hot Reserve
- Spinning Reserve

Also explain how these are related to the **economics of power generation.**

🔹 **Q6.** Define and explain the **Load Curve** in power generation.

◆ **Q1.** How is the **Load Curve useful in planning and operation of a power plant?**

◆ **Q2.** Differentiate between **Base Load Plant** and **Peak Load Plant.**

◆ **Q3.** Explain the following curves with neat diagrams:

- Daily Load Curve
- Load Duration Curve
- Integrated Load Duration Curve

◆ **Q4.** Explain the following terms:

a) Demand Factor
b) Average Demand
c) Maximum Demand
d) Plant Capacity Factor
e) Plant Use Factor
f) Diversity Factor
g) Load Factor and Plant Load Factor (with formula)
h) Energy Supplied per Year

◆ **Q5.** What is meant by **Load Curve**? Explain its **significance in power generation.**

✨ UNIT – 6: Basics of Transmission & Distribution ⚡ 📡

🔹 **Q1.** Explain the **AC Transmission and Distribution System.**

🔹 **Q2.** Explain the following **distribution systems:**

- Radial System
- Ring Main System
- Interconnected System

🔹 **Q3.** Write the **functions and importance of HVDC transmission.**

🔹 **Q4.** Explain the **components and working of an AC transmission and distribution system.** Also draw a **typical single line diagram of electric supply system.**

🔹 **Q5.** Explain the concept of **Black Start Restoration.** How is it done during a **blackout?** Also write the **steps involved.**

🔹 **Q6.** What are the **advantages of DC transmission?**

Q1. Define and explain the following terms in brief:

- (a) Voltage Level
- (b) Line Length
- (c) Types of Current (AC/DC)

Q2. Differentiate between a Substation and a Receiving Substation. State their primary functions.

Q3. Briefly explain the concept of Primary and Secondary Transmission lines.

Section B: Descriptive Questions (10 x 3 = 30 Marks)

Q4. High Voltage Transmission:
Write down the key characteristics and advantages of using high voltage for power transmission.

Q5. Substation Diagram (Case I):
Draw a detailed Single Line Diagram (SLD) of an **11 kV / 400 V substation** and explain its working.

Q6. Substation Diagram (Case II):
Draw and explain the Single Line Diagram (SLD) of a **66/11 kV substation.**

Q5. Substation Diagram (Case I):
Draw a detailed Single Line Diagram (SLD) of an **11 kV / 400 V substation** and explain its working.

Q6. Substation Diagram (Case II):
Draw and explain the Single Line Diagram (SLD) of a **66/11 kV substation.**

Section C: Long Answer/Application Questions (25 Marks)

Q7. Substation Components and Functions:
Explain the Single Line Diagram of a **66/11 kV substation** in detail. List and explain the functions of all major components used (e.g., Transformers, Circuit Breakers, Isolators, etc.).
